

# **MCSE699J Dissertation-II**

## **EVIDENCE GUIDED MULTIMODAL FAKE NEWS DETECTION USING TEXT – IMAGE CONSISTENCY AND EXTERNAL KNOWLEDGE**

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# ABSTRACT

The spread of false information via posts that combine textual assertions with visually appealing but potentially misleading photos has become a major social concern due to the rise of fake news on social media platforms. The majority of current methods for detecting fake news rely on supervised multimodal learning or text-only analysis, which frequently rely on erroneous correlations within datasets and lack explicit factual verification and interpretability. These restrictions limit their resilience and usefulness in situations when transparency and trust are crucial. This paper introduces an explainable multimodal false news detection system that combines retrieval-augmented evidence verification, visual–textual consistency analysis, and textual representation learning to overcome these issues. The suggested system uses a vision-language representation model to assess the semantic alignment between text and images, processes the textual and visual elements of a social media post independently, and uses a semantic retrieval mechanism to obtain pertinent external information from reliable sources. The model can determine if the claim is supported or refuted by outside factual information by using the retrieved evidence to calculate an evidence consistency score. A supervised learning model is used to classify these diverse signals into a uniform feature representation and generate a fake or real forecast along with interpretable evidence and a confidence score. When compared to traditional supervised baselines, experimental evaluation on a large-scale multimodal dataset shows that adding external evidence and cross-modal verification enhances the robustness and transparency of fake news identification. The suggested architecture, which offers a dependable and expandable solution for trustworthy false news identification in genuine social media contexts, places an emphasis on explainability and factual basis.

**Keywords:** Fake News Detection, Multimodal Learning, Retrieval-Augmented Reasoning, Image–Text Consistency, Evidence-Based Verification, Social Media Analysis

# TABLE OF CONTENTS

| Chapter No. | Contents                                              | Page No.  |
|-------------|-------------------------------------------------------|-----------|
|             | <b>Abstract</b>                                       | <b>i</b>  |
| <b>1.</b>   | <b>INTRODUCTION</b>                                   | <b>1</b>  |
|             | 1.1 BACKGROUND                                        | 1         |
|             | 1.2 MOTIVATIONS                                       | 2         |
|             | 1.3 SCOPE OF THE PROJECT                              | 3         |
| <b>2.</b>   | <b>PROJECT DESCRIPTION AND GOALS</b>                  | <b>4</b>  |
|             | 2.1 LITERATURE REVIEW                                 | 4         |
|             | 2.1.1 Traditional Text-Only Fake News Detection       | 4         |
|             | 2.1.2 Multimodal fake news detection                  | 5         |
|             | 2.1.3 Semi-supervised fake news detection             | 5         |
|             | 2.1.4 Graph-based and propagation- oriented detection | 6         |
|             | 2.1.5 LLM-Based Fake News Detection Systems           | 6         |
|             | 2.1.6 Retrieval-Augmented Fact-Checking Systems       | 7         |
|             | 2.2 GAPS IDENTIFIED                                   | 7         |
|             | 2.3 OBJECTIVES                                        | 8         |
|             | 2.4 PROBLEM STATEMENT                                 | 9         |
|             | 2.5 PROJECT PLAN                                      | 10        |
| <b>3.</b>   | <b>TECHNICAL SPECIFICATION *</b>                      | <b>12</b> |
|             | 3.2 FEASIBILITY STUDY                                 | 12        |
|             | 3.2.1 Technical Feasibility                           | 12        |
|             | 3.2.2 Social Feasibility                              | 12        |
|             | 3.3 SYSTEM SPECIFICATION                              | 12        |
|             | 3.3.1 Software Specification                          | 12        |
|             | 3.3.2 Hardware Specification                          | 13        |
| <b>4.</b>   | <b>DESIGN APPROACH AND DETAILS *</b>                  | <b>14</b> |
|             | 4.1 SYSTEM ARCHITECTURE                               | 14        |
|             | 4.2 DESIGN                                            | 16        |

|           |                                         |           |
|-----------|-----------------------------------------|-----------|
|           | 4.2.1 Data Flow Diagram                 | 16        |
|           | 4.2.2 Class Diagram                     | 17        |
| <b>5.</b> | <b>METHODOLOGY AND TESTING *</b>        | <b>19</b> |
|           | 5.1 Module Description                  | 19        |
|           | 5.2 Testing                             | 20        |
|           | 5.2.1 Data Collection and Preprocessing | 20        |
|           | 5.2.2 Evaluation Results                | 21        |
|           | <b>REFERENCES</b>                       | <b>24</b> |